TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	91.9 ± 0.83
Correlation Stability	95.4 ± 0.49
Distribution Consistency	94.7 ± 1.42
Entropy Preservation	95.8 ± 0.98
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	98.2 ± 0.87
Privacy Risk	11.9 ± 3.42
Data Density	100.0 ± 0.00
Coverage Score	44.3 ± 3.49
Feature Accuracy	95.0 ± 1.48
Novelty Score	0.8 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	43.234	-1.512	3.4%	OK
TB	3.299	3.879	+0.580	17.6%	Warn
DB	1.486	1.810	+0.324	21.8%	Warn
Alkphos	290.576	301.264	+10.688	3.7%	OK
Sgpt	80.714	74.994	-5.720	7.1%	OK
Sgot	109.911	105.039	-4.872	4.4%	OK
TP	6.483	6.571	+0.088	1.4%	OK
ALB	3.142	3.197	+0.055	1.7%	OK
A/G Ratio	0.947	0.957	+0.009	1.0%	OK
Selector	1.286	1.262	-0.025	1.9%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	91.9 ± 0.83
Correlation Stability (*2)	95.4 ± 0.49
Distribution Consistency (*3)	94.7 ± 1.42
Entropy Preservation (*4)	95.8 ± 0.98
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	98.2 ± 0.87
Privacy Risk (*7)	11.9 ± 3.42
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	44.3 ± 3.49

Feature Accuracy (*10)	95.0 ± 1.48
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.7042735042735042	0.6982905982905983	-0.007692307692307665	
F1	0.8128002697439015	0.8128343619099241	0.810461594887539	-0.002338674856362410 5	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6598290598290599	0.6658119658119659	0.6632478632478632	0.003418803418803418	
F1	0.7624676455734888	0.7694055087892642	0.7673435164113116	0.004875870837822793	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5487179487179488	0.5435897435897437	-0.01538461538461538	
F1	0.5587432119380917	0.5419699535878331	0.5330875564261168	-0.025655655511974913	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6982905982905983	0.7085470085470085	0.005982905982905995	
F1	0.7997903571549948	0.796294265040087	0.8029973272290742	0.003206970074079496	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7068376068376068	0.7068376068376068	0	
F1	0.8279091624529537	0.8279091624529537	0.8279091624529537	0	

Model: RF

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7102564102564103	0.7017094017094017	0.7256410256410256	0.01538461538461535	
F1	0.8123608174314491	0.8046682388341992	0.8217500698632587	0.009389252431809459	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6871794871794872	0.6829059829059829	-0.0034188034188034288	
F1	0.7971299319768493	0.7953130734872139	0.7914739774804669	-0.005655954496382576	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.688034188034188	0.6974358974358974	0.02307692307692308	
F1	0.7821343746683586	0.7944134951078649	0.8014636455185977	0.01932927085023909	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6829059829059829	0.7017094017094017	0.7	0.01709401709401709	
F1	0.7896120512268338	0.7991107089706302	0.7978769256366987	0.008264874409864852	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6991452991452991	0.6957264957264957	0.6974358974358974	-0.00170940170940171744	
F1	0.8055388887121644	0.8099917654931262	0.8090067214777955	0.003467832765631096	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	37	Male	0.7	0.2	176	22	19	7.3	4.6	1.66
2	45	Male	0.6	0.2	170	15	15	6.2	3.1	0.9
3	37	Female	0.9	0.1	165	24	28	7.3	4.7	1.5
4	46	Male	15.8	7.0	450	38	88	5.7	2.1	0.5
5	35	Female	0.7	0.2	279	40	30	7.9	3.6	0.9
6	65	Male	0.8	0.1	165	22	25	5.8	3.0	1.0
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	1.0	0.3	216	55	58	7.6	4.3	1.34
2	45	Male	2.3	1.3	163	54	38	8.5	5.0	1.7
3	62	Male	7.3	3.6	298	220	367	7.3	3.3	0.8
4	20	Female	0.7	0.2	186	78	48	6.1	3.9	1.6
5	17	Male	0.7	0.1	179	28	34	7.2	4.0	1.2
6	60	Male	0.6	0.2	187	16	23	5.9	3.2	1.18
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	75	Male	1.4	0.4	152	15	24	8.1	3.0	0.6
2	33	Male	0.7	0.2	168	21	25	5.9	3.1	1.1
3	58	Male	1.4	0.4	182	13	26	7.0	3.4	0.9
4	31	Female	0.7	0.2	165	21	30	7.4	3.2	0.6
5	44	Male	2.2	1.6	282	132	68	7.8	4.0	1.1
6	33	Male	3.4	1.6	206	1680	850	7.1	3.6	1.1
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28	Male	1.0	0.2	210	20	19	8.0	4.0	1.0
2	50	Male	0.8	0.2	175	55	95	5.5	2.2	0.6
3	46	Male	0.6	0.1	196	155	202	7.5	3.9	1.0
4	36	Male	1.6	1.0	180	18	26	8.0	4.0	1.0
5	20	Male	0.7	0.2	191	37	33	6.6	3.6	1.2
6	8	Female	0.7	0.1	460	28	23	7.9	3.9	0.9
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	11.1	6.0	298	25	19	5.4	2.7	1.2
2	65	Male	0.8	0.1	148	20	25	5.8	2.7	0.8
3	37	Male	1.7	1.0	342	28	50	5.9	2.8	0.8
4	60	Male	5.7	2.5	168	220	245	6.8	3.0	0.7
5	50	Male	1.2	0.4	310	36	42	8.2	4.0	1.0
6	35	Male	0.8	0.2	158	29	34	7.4	3.1	0.7
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Male	0.6	0.2	243	21	43	8.6	4.0	1.0
2	58	Female	0.7	0.2	298	29	90	6.7	3.1	0.7
3	74	Female	0.9	0.2	162	23	20	7.9	4.1	1.0
4	28	Female	0.9	0.2	309	131	140	5.4	3.0	1.1
5	73	Male	0.9	0.3	215	16	23	5.8	2.9	0.8
6	37	Male	2.8	1.3	105	25	41	5.6	2.5	0.8
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	36	Male	0.8	0.2	190	31	39	7.0	3.1	0.6
2	50	Male	0.6	0.1	218	17	20	5.0	2.9	1.4
3	75	Male	2.9	1.3	260	27	22	5.5	2.4	0.7
4	23	Male	1.0	0.3	75	15	26	6.4	3.1	1.0
5	58	Female	0.7	0.1	155	37	59	6.1	3.2	1.0
6	75	Male	2.5	1.2	375	42	66	6.4	3.2	1.0
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	34	Male	4.1	2.2	298	950	1500	6.6	3.8	1.2
2	63	Male	0.9	0.2	187	28	14	5.5	3.1	1.1
3	45	Male	0.6	0.2	196	44	18	8.1	4.4	1.2
4	39	Male	0.6	0.1	147	28	34	2.8	1.6	1.72
5	60	Male	8.9	4.0	388	233	602	7.4	3.0	0.6
6	12	Male	0.9	0.2	401	30	29	7.1	4.0	1.3
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	27	Male	1.2	0.5	179	20	24	8.0	4.4	1.3
2	75	Male	10.6	4.0	386	43	41	6.1	2.8	0.8
3	42	Male	30.5	14.1	214	119	540	6.8	2.7	0.9
4	60	Male	6.8	3.2	204	321	500	5.4	3.0	1.9
5	55	Male	3.3	1.8	171	42	71	7.5	2.9	0.6
6	12	Male	0.5	0.1	182	35	41	4.8	2.0	0.7
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	70	Male	0.9	0.3	170	16	19	6.7	3.7	1.2
2	42	Male	1.0	0.3	182	18	19	6.7	3.8	1.4
3	51	Male	4.0	2.1	230	13	29	6.7	2.8	0.9
4	43	Male	2.7	1.0	554	407	850	6.4	3.1	0.8
5	42	Female	0.5	0.1	188	52	38	5.0	1.8	0.6
6	11	Male	0.8	0.1	348	30	82	6.1	3.7	1.6
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3